

Miles Jennings

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EDUCATION

University of California Irvine

Irvine, CA

B.S Computer Engineering — Deans Honor List

Sept. 2022 – June 2026

Relevant Coursework: Organization of Digital Computers (Computer Architecture) Electronics I, Electronics II, Network Analysis I, Network Analysis II, Principles of Digital Design, Processor Hardware/Software Interfaces, Organization of Digital Computers

Organizations: HyperXite, FUSION, FASAE

EXPERIENCE

Test Engineer Intern

Buena Park, California

Leach International Corp.

March 2026 – Present

- Troubleshoot, made documentation for, and repaired test equipment by analyzing existing system architectures and adapting to legacy setups, eliminating production bottlenecks and ensuring on-time delivery
- **Developed** a **Python**-based capacitor test/conditioning system (LCR6002, High Voltage Power Supply, Safety Analyzer) for validation of **490** components, eliminating outsourced testing and reducing related company costs by **60%**
- Built and maintained hardware test setups using **oscilloscopes**, **multimeters**, and **power supplies** to perform functional and performance testing on electrical components

Control Systems Engineer

Irvine, California

HyperXite – SpaceX Hyperloop

Aug. 2025 – June 2026

- Designed, tested, and optimized STM32H753ZI-based control hardware in **C/C++**, interfacing with sensors via **UART**, **I2C**, and **CAN** while evaluating components for performance and cost improvements
- Built ESP32-based bidirectional communication system using ESP-NOW for real-time control and telemetry between pod and station and utilizing FreeRTOS to schedule critical tasks
- Designed a custom control board in **Altium** to integrate the pod's control hardware, collaborating with cross-functional teams to ensure reliable power distribution and acquisition of critical sensor and telemetry data

P2S Technology Intern

Long Beach, California

P2S Engineering

June 2025 – March 2026

- Assisted in the design and documentation of low-voltage and IT infrastructure systems for large scale buildings
- Improved design accuracy and project timelines by reducing regulatory revision cycles by **11%** through proactive 3D modeling and cross-functional coordination
- Collaborated with multidisciplinary teams to coordinate technology system requirements with electrical and mechanical designs to ensure reliable infrastructure design

PROJECTS

MIPS & Pipeline Processor | Verilog

Jan. 2025 – Feb. 2025

- Designed and implemented a **Verilog**-based MIPS processor with both single-cycle and pipelined architectures
- Built key datapath and control components, including **ALU controllers**, **multiplexers**, register file interfaces, and combinational control logic
- Validated processor functionality through simulation, debugging instruction execution, control signal generation, and pipeline stage interactions

4-Bit Full Adder/Subtractor | Cadence, Spectre, CMOS logic

November 2025

- Designed and implemented a 4-bit full adder in **Cadence** using custom transistor-level schematics and layout
- Utilized Spectre to perform simulations and ran DRC/LVS verifications to test functionality, timing behavior, and power/performance characteristics
- Iterated layout and schematic revisions based on verification results to ensure manufacturability and compliance with design rules

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Verilog, RISC-V, MIPS

Software/Tools: LabVIEW, Revit, AutoCAD, KiCad, Bluebeam, Arduino IDE, VSCode, Vivado, LTSpice, Cadence, Linux, Git, GitHub, STM32CubeIDE, MobaXterm

Hardware: Oscilloscope, Power Supply, Multimeter, Circuit Design & Analysis, Microcontrollers, ESP32, Embedded Systems, PCB Design, Arduino, Power Supplies, FPGA